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Civil Infrastructure Damage Classification using YOLO

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Abstract

Roads are necessary for daily life because they provide transportation between various places. Road maintenance ensures that they remain safe and operational. On the road, there are a lot of potholes, which can lead to more accidents. Around 3,597 people die as a cause of potholes each year. Potholes can be avoided by detecting them and reporting them to the appropriate authority. The objective of this project was to create a pothole detection system that was specifically designed to find potholes. This project analyzed and implemented using YOLOv4 designs to locate potholes. The YOLO (You Only Look Once) method was used to construct a model that can instantaneously recognize objects after being trained to look for potholes in YOLOv4. The real-time object recognition technology detects the pits, which are presented in real-time and marked with a box. This technique provides accuracy between 80% and 95%. YOLOv4 was used to annotate and train on a collection of pothole picture data, and the results were graded for recall, accuracy, and mAP. A variety of indentation-based photos were used to validate the model.

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1. Introduction

A great deal of research has previously been done on obstacle identification, and several strategies for avoiding various impediments in various situations have been investigated. However, autonomous agents avoiding impediments, which simply comprised extruding barriers, has received the majority of attention.

“Roads play a key role in road transportation by connecting remote towns, rural areas on the interior, and hills without rail access. The railroads are free to use. The third-largest road network in the world is in India. The Nagpur project of 1943 separated the roads into four groups:

- National Highway
- Highway
- State Highway
- District and Village Roads

National highways are under the National Highway Authority of India's (NHAI) management (Omar & Kumar, 2020a).



Figure 1: Road condition with pothole

As a result, detection techniques have become highly system-specific and unsuitable for wider applications. Road transport was relatively easy and cheap, although the comfort of the trip depends on the quality of the roads. Potholes are a major source of frustration for drivers. Heavy traffic and potholes from heavy rain can cause accidents and even death.

As a result, potholes are a major concern for drivers, putting them at risk of accidents and vehicle damage. Unexpected bumps and grooves can lead to more accidents. Potholes should therefore be repaired to improve ride comfort and minimize possible risks.(Chavan et al., 2022a)

The intelligent transport system "ITS" has evolved into a digital society and emerged as a central element in building smart cities that make the environment of residents safer in all respects. Whether people move for work, school, college, or for other reasons, it is a major concern for all cities. To move around cities, people use mass transit systems. Combined with the intelligent transportation system "ITS", it saves time, makes cities smarter and transports people. ITS aims to alleviate traffic congestion and increase transport efficiency. Today, traffic can be managed with sophisticated and flexible devices. The Intelligent Transport System (ITS for short) is responsible for most of the traffic. Intelligent Transport System "ITS" this is a computer-assisted method of using a suite of automotive transport apps to review real-time feeds from highly sensitive roadside smartphone sensors and devices, as well as cameras mounted on signal poles. Smartphones are equipped with state-of-the-art CPUs and various advanced built-in sensors(Asad et al., 2022).

The state of the roads must be determined to decide which routes require rehabilitation or upkeep. This activity can be carried out either manually or automatically. Currently, these examinations are performed manually in Spain, which is both costly and time consuming because a human specialist is required to assess the state of these roads (H.-W. Wang et al., 2015a). Expert decision-making is defective and can provide erroneous information regarding the ultimate status of a road section since it is reliant on individual subjective judgements based on false impressions. The advancement of science and technology, as well as the reduction in the production costs of electronic components, allows for the creation of new approaches and instruments for objective automated detection. As a result, universities and research institutions collaborate to produce clear photographs of roads and detect various types of degradation in transportation routes (Das et al., 2020).

Below figure displays the annual quantity of potholes patched in San Antonio, Texas, from the fiscal year 2013 through the fiscal year 2021. COVID-19 patched roughly 100,520 potholes in 2019 and approximately 80,937 potholes in 2021, according to the San Antonio Pothole Patrol Team. In Chicago alone, more than 156,000 holes will be filled by 2021. These figures are considerably high in cities. The United States has an extensive road network. According to the Bureau of Transportation Statistics, the United States has around 746,100 miles of road. A journey of this length is difficult, if not impossible, to follow with simply human effort. Automatic pothole detection technology is built into the automobile to help drivers discover potholes, alert them when required, and schedule repairs (Chang et al., 2004).

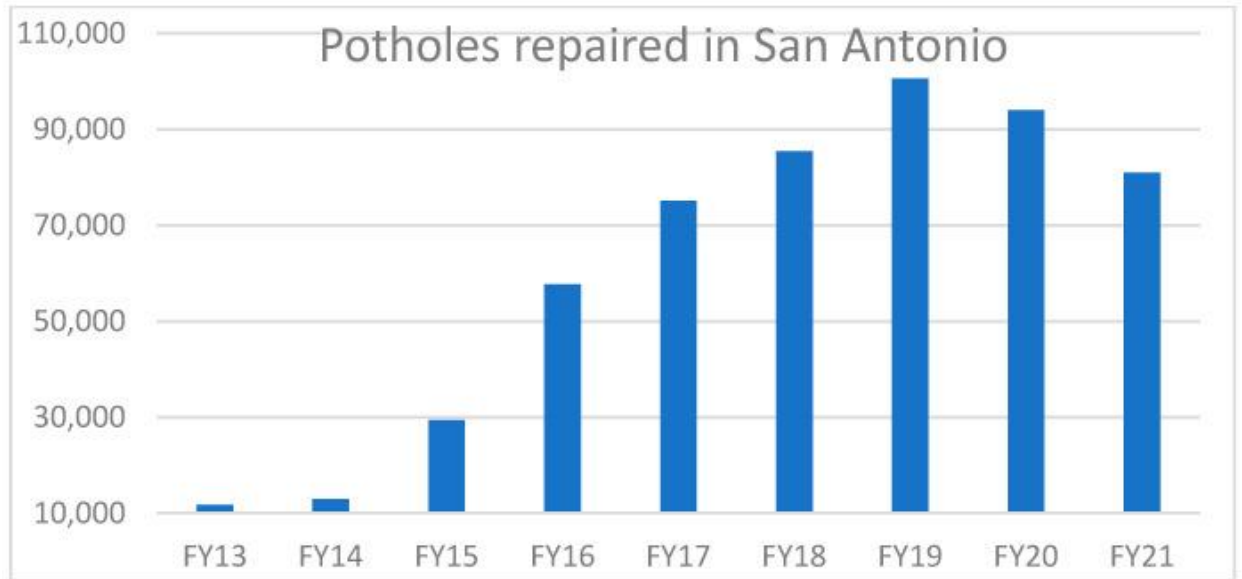


Figure 2: Using statistics from FY13 through FY21, the quantity of potholes patched year in San Antonio, Texas, in the United States (Chang et al., 2004)

Potholes are detected using vibration sensors in sensor-based technologies. False negatives and false positives can reduce pothole identification accuracy because the vibration sensor views road gaps as potholes rather than a crater in the center of the road. 3D reconstruction technology captures 3D road data to detect potholes. Misplacement of the camera can reduce the effectiveness of resurfacing efforts and requires expensive setup and computation. The accuracy of conventional image processing methods for detecting potholes in the road is quite high, but these methods require more complex operations, such as manual extraction of features and replacement change parameters and image processing steps for different road conditions (PMC8704745, n.d.).

Smartphones are mobile devices with sensors like gyroscopes, GPS, microphones, cameras, and accelerometers. ITS is implemented and used in developing countries for real-time navigation, lane enforcement, travel time prediction, and traffic warnings. The efficiency, capacity, and safety of the road transportation system should all be improved. ITS is widely used in several developing nations. While ITS implementations vary between far-off nations, the objective is constant. This suggests more effective transportation systems, reduced traffic, more security, and improved traveler comfort. Manually analyzing and interpreting visual road surface data is a time-consuming and expensive process, with evaluation subjectivity and experience heavily influencing results. The main purpose of this system is to identify road potholes using deep learning technology. This automates the process of identifying potholes, making them easier to spot when approaching the road. The authorities were concerned about the issue of asphalt pavement to avoid unfavorable eventualities. These sidewalks have two major pavement faults, cracks, and potholes, which are vulnerable to reasons such as traffic,

weather, ageing, a lack of building materials, and insufficient drainage systems. A pothole is depicted in Figure 3 as a dip in the road's surface. (Ng et al., 2019)



Figure 3: pothole on roads (Ng et al., 2019)

Potholes may lead to crashes, unpleasant driving experiences, automotive troubles, and other calamitous outcomes. Pothole repairs should be prioritized to decrease their influence on adverse events. According to the World Health Organization, road accidents will be the sixth leading cause of death by 2030.

You Only Look Once is an abbreviation for the phrase You Only Look Once. It is a one-shot detector, which means that the input picture is only sampled once, hence the name. It is one of the most extensively used object identification algorithms and has gained much attention since its first publication. It is extremely quick and exceeds other current object detectors. It has several uses, including self-driving automobiles, video analytics, and object recognition. Pothole detection is included. Over the years, many variations of the YOLO algorithm have been developed. They all build on prior versions and are considerably improved over the previous versions. YOLOX is the most recent reported object identification model.

Algorithms have been tuned using a variety of approaches to enhance both speed and accuracy. The YOLO method is one of the most extensively used object identification models for pothole detection. As a result, the goal of this work is to train and assess YOLOv4 algorithm for detecting road potholes. The section that follows, offers an overview of some of the most widely utilized YOLO pothole detecting methods.

1.1 Motivation

A growing world population puts more strain on infrastructure and flooded roads. The main motivation for creating vehicles intelligent enough to assist drivers in many aspects is the increasing difficulty of managing vehicular traffic.

One of the serious problems with roads is due to various factors such as rain, oil spills, traffic accidents, etc. One of the most typical kinds of damage to roadways is potholes. It

happens on roadways as a result of a variety of variables, including human actions and natural elements like snow and rain. Since pothole creation is influenced by a variety of causes, it is impossible to forecast where they will appear. Considering that India has almost 5 million kilometers of roads, locating and fixing potholes is an extremely difficult undertaking. Because professional road inspection requires a lot of time and resources, it takes the government a long time to find road problems. To keep roads in good condition, significant sum of money is invested annually (Park et al., 2021).

Numerous research has been published to automate the process of finding potholes on the road. The most widely used methods for finding potholes in the road are (i) Vibration Technology, which uses accelerometers and other sensors to detect bumps in the road as a vehicle passes over it; (ii) Laser Scanning, Stereo Vision, or Dynamic Sensor methods; (iii) Image Processing methods using morphological techniques like GLCM (Gray-Level Co-Occurrence Matrix) and RBF (Radial Basis Function); and (iv) using Artificial Intelligence, computer vision engineering (H.-W. Wang et al., 2015b).

1.2 Deep Learning and Detection system

Deep learning (DL) principles have recently gained interest from academics and entrepreneurs due to their ability to tackle classification problems like, understand dimensionality reduction, recognize objects, and interpret spoken languages. DL uses the principle of multi-layer neural networks to extract the inherent quality of data from lowest to highest level. They can describe a large amount of structure in your data, so you can look at the actual data and make decisions. Most ITS (Intelligent Transportation System) departments and researchers in this field are also considering developing self-driving vehicles to make ITS cost-effective and minimize human life risk. This approach has the added benefit of saving time. In recent decades, a great deal of importance has been placed on safe automated systems, data-driven driving through the traffic sign recognition system, autonomous cars, and driver assistance system (Harikrishnan & Gopi, 2017a).

The ultimate goal of object detection is to find the elements of interest surround them with a rectangular bounding box and classify each one. Object detection is used in a variety of fields, from crop monitoring to real-time ball tracking in sports to pedestrian detection in self-driving cars. The object localization issue, which seeks to both identify and pinpoint the positions of image objects is the first significant obstacle to object identification. Researchers usually use a multi-task loss function to address this issue since it penalizes both classification errors and localization errors (Silva et al., 2020).

Region-based CNNs represent a diverse set of object detection frameworks. To improve it employs CNN processing. Researchers developed fast R-CNN to improve preliminary R-CNN findings. Fast R-CNN provides not only impressive speedups but also accuracy since the classification and localization tasks are enhanced using a combined multitasking

loss function. Each candidate region holding a possible item is compared to the actual thing in the image. The candidate regions are then penalized for misalignment of the bounding box as well as misclassification. As a result, two types of terms comprise the loss function (Harikrishnan & Gopi, 2017b),

$$\mathcal{L}(p, u, t^u, v) = \overbrace{\mathcal{L}_c(p, u)}^{\text{classification}} + \lambda \overbrace{[u \geq 1] \mathcal{L}_l(t^u, v)}^{\text{localization}},$$

where “the localization term is the smooth L1 loss of the four rectangular location components and the classification term is a logarithmic loss imposed on the projected probability of the true object class ‘u’. If the item does not exist i.e., (u=0), the background class is not penalized for localization”. You may also adjust the selection to heavily favor taxonomy or localization. The suggested technology makes accurate remote detection of potholes possible. Figure 4 depicts the situation when certain wells are unrecognizable from the LR picture. Experiment findings reveal that when a cutting-edge object detector is trained on this SR pictures, it surpasses “detectors trained on LR images” in terms of accuracy and capacity to detect small gaps in frames. It is hoped that this work will increase industry awareness of the potential application of using HR imagery to identify road potholes (Harikrishnan & Gopi, 2017b).

Image processing techniques employ various statistical methodologies, such as GLCM, RBF and others to extract and distinguish distinct textures and features of images. These approaches are commonly used in combination with other machine learning methodologies due to the large amount of computing power required. The GLCM image preprocessing method analyses and separates textures while also taking into consideration the spatial relationships between pixels. Radial Basis Functions (RBFs) are real-valued functions whose primary role is to linearly combine multivariate functions. It is an ANN (Artificial Neural Network) capable of changing the form of the input signal. The phrase "morphological image processing" refers to a class of image processing methods that concentrate on the form of individual features in grayscale pictures (Hoang, 2018).



Figure 4: The first image shows the instances found in the LR image and the second image shows the instances found in the SR image showing more occurrences of pits, especially those in the distance (Harikrishnan & Gopi, 2017b).

1.3 Objectives

- The primary objective of this project is to create a system that can detect potholes using image and video.
- To detect pothole using YOLOv4 algorithm.

1.4 Research Question

The research questions were developed to address the main issue in the project. Few research questions are as follow:

- Why Detection of pothole needed?
- How YOLOv4 algorithm detect pothole?

1.5 Research Structure

The structure of the project going forward is as follows:

- Section 2 is the literature review on picture super-resolution, pothole detection methods, and studies combining super-resolution with object recognition for tiny item recognition tasks.
- In section 3 YOLOv4 along with the approach that is being suggested by describing the datasets used to train and test the algorithm.
- Section 4 gives experimental details, the comparative performance assessment findings from trials with various datasets are shown.
- Section 5 serves as our final section or the conclusion.

2. Literature Review

This chapter's primary goals are to first assess the significance of the general study field and then to suggest prospective areas for fresh contributions. The majority of the chapter is devoted to a critical assessment of the methodologies employed in the field in order to choose the most effective technique to address the research topic (J et al., 2020a).

Identifying potholes is a method of identifying road anomalies that have formed for a variety of reasons, including increased rainfall, poor road maintenance, natural disasters, etc. “One of the biggest problems in most developing countries is road deterioration. This is due to poor road maintenance, frequent rainfall, heavy traffic, rough terrain, natural disasters, poor choice of road construction materials, etc.” Below are studies conducted by multiple researchers to identify road damage such as potholes, cracks, and road deterioration, etc. (Ping et al., 2020a)

Researchers suggested a successful DL method that can automatically recognize potholes on lane surfaces. The “YOLOv3, SSD, HOG with SVM, and quicker R-CNN with preprocessed datasets models” are among the four that are trained and tested. The paper described methods for enhancing road damage databases, considering how the enhanced data influences the precision of identification and classification. Each class chooses one of five strengthening exercises. Using an effective technique for data augmentation, a perfect training dataset is produced (Sirohi et al., 2020).

The authors give a broad overview and a thorough investigation of the use of deep learning models in intelligent transportation systems. Let's get into more depth regarding the development of ITS deep learning research. According to the researchers, numerous CNN model types were applied in their most current ITS for pedestrian identification, traffic light recognition, vehicle categorization, and traffic sign recognition. In their study a straightforward yet efficient method for detecting road damage is suggested (Y. J. Wang et al., 2018a).

The authors employed cutting-edge deep object identification models, SSD, and quicker RCNN to solve this problem. According to experimental findings, the suggested model

wins the IEEE Big Data Road Harm Detection Competition and provides accurate detection outcomes. The author created a new dataset called India Driving Dataset to examine the issue of autonomous navigation in unstructured driving conditions using ResNet101 and PANet with his MaskRCNN. In their study, scientists presented a simple semantic segmentation technique that uses high-speed photography to identify areas of damaged road surfaces. They used two speed metrics and four accuracy measures to assess the effectiveness of the proposed method (Y. J. Wang et al., 2018b).

A “unique method for rapid damage inspection and traffic sign recognition in unstructured environments is proposed in a paper based on mobile laser scanning (MLS) data. We validate this approach using the point cloud of the RIEGL VMX-450 MLS system. The Indian Road system does not follow sufficient guidelines for bump design and maintenance, so uneven bumps caused by body roll can damage vehicles and harm the health of drivers. In their study, the researchers used deep learning techniques to detect and warn drivers in real-time before approaching unmarked speed bumps and used stereo vision techniques to detect vehicle movements from them. They presented a system for estimating distance”(Y. J. Wang et al., 2018c).

2.1 Review of pothole Detection Methods

Object recognition, a critical area in data science, which has already shown astounding achievements. Face recognition, counting things in a picture or room, and tracking moving objects are all possible with object detection. Pothole identification is one of many practical uses of object detection that have long been a source of frustration for motorists, local and state governments. Weather particularly winter weather, damage roads and automobiles which causes accidents and can be costly to repair.

With the present pandemic threat, the globe is quickly heading towards an autonomous environment, which has become urgently necessary. Numerous businesses, including road building and maintenance, have been negatively impacted by the outbreak. A safe working environment for employees is a primary responsibility for road repair in these difficult circumstances. To some extent, autonomous systems designed to reduce reliance on humans that can help with this. This work suggests one such system, a pothole detection and size estimating system (Y. J. Wang et al., 2018c).

Deep learning approaches attempt to learn feature hierarchies that consist of low-level features mixed with high-level features. By learning features at multiple levels of abstraction, systems can learn sophisticated features that directly change inputs from data to outputs, rather than relying on human-generated features. Deep learning approaches have proven useful in a variety of computer vision applications such as image segmentation, object detection, and classification (Chang et al., 2004).

This paper has created an effective “CNN model in order to efficiently identify potholes in real time. They come in various road conditions, varied forms, and different sizes in the trials carried out in this paper, which used a collection of pothole photographs recorded under different lighting circumstances. Record of potholes trained on 10 distinct CNNs, including a Faster R-CNN with five different backbones, two versions of YOLOR, and three YOLOv5 variants (Yl, Ym, and Ys) (Proposed CNN called ResNet50, VGG16, MobileNetV2, InceptionV3, and MVGG16). 91.9% accuracy was recorded by Faster R-CNN ResNet50, which was followed by Ym and Yl, planned MVGG16, and lastly MobileNetV2.”

In image processing, object detectors rely on manual representation and eliminate low-level features. To identifying pits/potholes in a single image or frame, several studies on image processing and other video-based techniques have been conducted thus far. Suggested to find potholes and tally how many there are in each frame, the author gathers a variety of frames, blurs the grayscale pictures using edge detection and morphological techniques, and then uses the Hough transform algorithm to extract features to determine colors (Bochkovskiy et al., 2020).

2D color photos were subjected to a morphological reconstruction method and a fuzzy C-means clustering approach to find depressions in the asphalt pavement. Furthermore, image processing to locate road potholes and exclude obtrusive elements like trees and cars, etc. Images are processed using simple image processing techniques such as canny filters and edge detection to identify potholes. The accuracy of the experiment was 81.8% and the recovery was 74.4%. The accuracy value of the test image is satisfactory, but not all line types give equal accuracy when the same technique is used. The authors detect potholes in all three steps (1) preprocessing to extract shadows from grayscale images and (2) extract candidates, find vanishing points to create a virtual path and (3) extract the amusement area cascading detectors with a certain threshold. Accuracy 88% is achieved with this technique and 71% recovery rate. Similarly, the authors detected potholes in three stages. (1) segmentation using histogram and morphometric filter to extract dark areas, (2) candidate regions are extracted based on various characteristics such as size and compactness and make decisions about them. (3) Candidate areas are pits by comparing pits and background features. The accuracy of pothole detection by image processing is affected by road conditions, such as the presence or absence of dirt on the road and changes in pothole size. Therefore, these approaches require adjusting the image processing parameters and processes for different road conditions, which is a tedious task. Furthermore, these methods are not suitable for real-time pothole detection because they require high computational power due to their computational complexity (Ahmed, 2021).

“Roads are connecting lines between different places and are used every day. Regular maintenance of roads keeps them safe and functional. Recognizing and reporting potholes

to the responsible authorities can help to eliminate them. This study was implemented and tested on different deep learning architectures to detect potholes. The images used for the training were collected with a mobile phone attached to the car's windshield, in addition to many images downloaded from the Internet to increase the size and variability of the database. Second, multiple object detection algorithms are used and compared to detect potholes in real time, such as SDD-TensorFlow, YOLOv3Darknet53 and YOLOv4Darknet53. YOLOv4 achieved the best performance with 81% recall, 85% accuracy, and 85.39% average accuracy (mAP). The processing speed was 20 frames per second. The system could detect potholes from a distance of 100 meters from the camera. The system can increase driver safety and improve the performance of autonomous vehicles by detecting upcoming potholes” (Chavan et al., 2022b).

“To recognize sidewalks and street holes, the proposed technique uses convolutional neural networks. This white paper provides a realistic method for pothole identification in the navigation module of the Sound of Vision Lite (SoV Lite) project based on previously published research. Using the collected datasets, YOLOv5, Mobilenet V1 and Mobilenet V2 Lite were trained on Nvidia RTX and tested on Nvidia Jetson NX. Due to mobile platform restrictions, Mobilenet V1 was chosen for integration into SoV Lite. Not only does the closer the item, the greater the likelihood of detection, but the visually impaired person in the system is concerned with local risks, thus we restrict the detection of negative barriers to a range of 8m in practice. They used internally collected frames from the ZED 2 stereo camera and public data for training and experimentation and annotated them for the specific objective of recognizing potholes and ditches in sidewalks and roadways. The resultant dataset was expanded and made public”.

Image classification by Russakovsky et al., Figure 5(A) predict the class of one or more objects in an image. Locating one or more elements in an image and creating a bounding box around their extent is called object localization. Figure 5(B) shows the combination of these two tasks. One or more elements in the image are identified and classified. It shows how segmentation assigns a label or score to each pixel in the image.

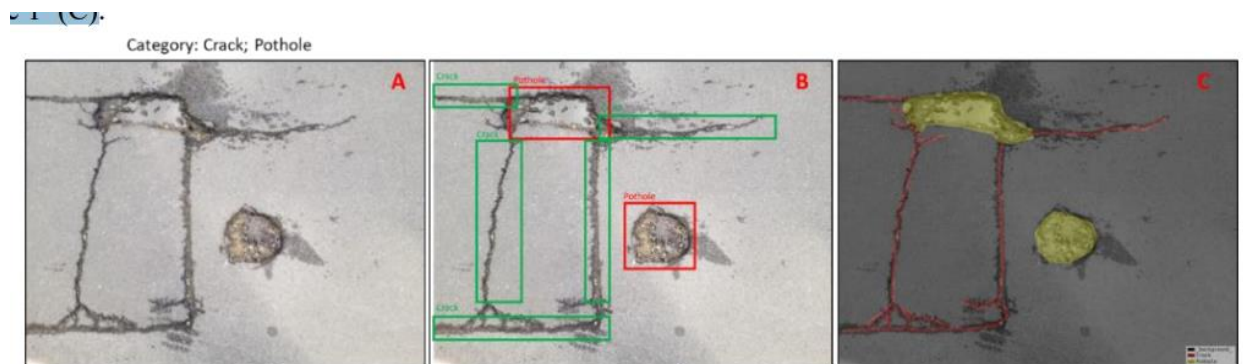


Figure 5: Identifying cracks and potholes using various ways

(C) Object identification techniques are typically split into two groups: Deep learning-learning and machine-learning based. It is important for machine learning techniques to first detect features using one of the methods described below, and then classify them using a method such as support vector machines. Deep learning algorithms, on the other hand, which are typically based on convolutional neural networks, can perform end-to-end object detection without explicitly specifying properties (CNN)(Liu et al., 2021).

In this application, integrated neural network (CNN) based algorithms are more commonly used to extract cavity features from images. On a given image, these models can accurately represent the non-linearity of the sample and perform automatic feature extraction. Due to its ability to filter out background noise and low contrast, it is also suitable for road shots. CNNs have been used effectively in many applications, but they are not useful in all situations. For example, if the item being recognized is small compared to the image, or if a high-resolution photo is used for remediation, given this problem, the computations required to analyze the data can be very expensive. This is due to CNN's high processing time and memory requirements. The first solution is a two-step approach that uses a location network (LCNN) to find individual potholes in an image and then a section-based classification network (PCNN) to identify classes (Darapaneni et al., 2021a).

In this application, a convolutional neural network (CNN) based technique is used to extract void characteristics from pictures. These models accurately describe pattern nonlinearities and can extract features from images automatically. Its low-contrast ruggedness and background noise filtering make it excellent for street photography as well. CNNs are beneficial in many applications, however they are not in every scenario. If the items to be detected are small in comparison to the image, or if high-resolution images are employed to address this problem, the computations necessary to interpret the data might be prohibitively expensive. This is because CNN requires a lot of memory and processing time. This application typically uses convolutional neural network (CNN)-based algorithms to extract gap features from photos. These models can correctly represent pattern nonlinearities in a given image and perform automatic feature extraction. It's also great for street photography thanks to its strong filtering of background noise and low contrast. CNNs have been used effectively in a variety of applications, but they are not useful in all situations. For example, if the elements to be recognized are small compared to the image, or if high resolution photographs are used to solve this problem, the computations required to analyze the data can be very expensive. Civil society experts have stressed the importance of potholes. Manual inspection of potholes in developing countries is largely based on human experience and can lead to misjudgments. These manual inspection methods require labor which is time consuming and expensive. A variety of techniques such as thermal imaging, vibration sensing,

computer vision, and scanning with 3D reconstruction can be used to locate the pits(Alfarrarjeh et al., 2018).

A sensor-based network called "BusNet" has been proposed to monitor road conditions, as opposed to cameras on public transport buses. It uses a variety of cheap, fast, and sensitive sensors and GPS. This technique is not perfect in all situations due to environmental conditions that can damage the sensors and affect BusNet functionality. With the availability of accessible, affordable, and practical cameras, computer vision and image processing-based approaches have gained popularity in recent years, replacing traditional manual inspection methods for identifying potholes (Darapaneni et al., 2021b).

The difficulty and cost of identifying potholes has been reduced by combining machine learning and deep learning methods. Arbawa and his colleagues have developed a system that detects vulnerabilities in the road. He studied the three qualities of contrast, correlation, and difference. The results showed that the combination of contrast and difference features gave better results, with an accuracy of 92.033% and a computation time of 0.0704 seconds for each image (Arbawa et al., 2021).

Oche et al, “used data gathered from cellphones and driving routes to examine five binary classification models (Naive Bayes, Logistic Regression, SVM, K-Nearest Neighbors (KNN), and Random Forest Tree) to compare various machine learning approaches. On the test set, Random Forest Tree and ANN had the greatest accuracy (0.8889). They adjusted the hyperparameters, bringing the random forest tree's accuracy up to 0.9444. When applied to varied root and extrapolation data, this model yields encouraging results. Al Pinget approach for detecting potholes combines machine learning and deep learning models”(Egaji et al., 2021).

Ye et al, “described two convolutional neural networks (conventional CNN and Prepon CNN). As a foundation for systems that investigate potholes and road defects. When the CNN was pre-cooled, the model which was trained on 96,000 digital street images reached a high identification accuracy of 98.95%. They determined that precooled CNNs are more robust and resilient under real-world situations than standard image processing approaches. K-Means and Sobel edge detection algorithms performed worse than CNN in terms of pit identification and position accuracy” (Ye et al., 2021).

Three different road types such as Pavement, Dirt, and Tarmac are classified using three different sets of data. For each type of road, further categories potholes. Finally, the CNN model trains on 7,000 photos from the RTK, KITTI, and caRINA datasets. They also suggested an app that alerts automobiles and hikers about imminent potholes on their paths. To find potholes and categories different types of roads, Wang and his associates employed YOLOv3 and a CNN model (Agrawal et al., 2021a).

2.2 Yolo Architecture

“Authors Joseph Redmon and Ali Farhadi released the initial version of YOLO in 2015. It is said to be one of the earliest object detectors to be classified as a single-shot detector. A single branching convolutional neural network predicts bounding boxes and class probabilities directly from frames in this case. Because the whole detection process is one network, algorithms may directly optimize end-to-end detection performance”. YOLOv1 was among the first object detection frameworks to locate objects without the use of anchor boxes. To predict the bounding box, this anchorless mechanism strictly considers points near the object's center (J et al., 2020b). This YOLO basic model was capable of real-time picture processing at 45 frames per second. It is significantly quicker than other detection approaches like DPM (deformable component model) or R-CNN. However, because it only uses midpoints for bounding box prediction, YOLOv1 has more localization mistakes and a substantially lower recognition rate due to the lack of anchor fields. Furthermore, because the complete image is collected, predicting tiny or dense items in the image becomes extremely challenging. This makes locating clusters of small and many craters more challenging. In 2017, YOLOv2 was published, which improved the training and performance of YOLOv1. Authors Joseph Redmon and Ali Farhadi published the book because YOLOv1 model lack of anchor boxes led to its poor accuracy, and YOLOv2 model employs anchor boxes in convolutional layers to achieve high localization accuracy and forecast k-mean clustering. This model includes high and low-level feature maps as well as fine-grained features that are highly beneficial for detecting tiny objects. YOLOv2 in VOC 2007 was able to attain 76.8 mAP at 67 FPS. This was sufficient to surpass other cutting-edge approaches like Quicker R-CNN, ResNet, and SSD, and the execution performance was still much faster (Lin et al., 2021).

For class labeling, YOLOv2 uses Darknet19. This is a network consisting of 19 convolutional layers, 5 max pooling layers and a final softmax layer. By taking input images of size 448,448 from 224,224, the overall mAP of YOLOv2 increased by ~4%. In YOLOv2, anchor boxes are responsible for localization, prediction, and bounding box prediction. Because of its fine-grained characteristics, YOLOv2 is better able to discern minute details. In order to forecast class probability or objectivity ratings, YOLOv3 does not employ softmax layers. Instead, logistic regression is employed. A feature pyramid network is used by YOLOv3 to detect objects of various sizes. Darknet-53 was later employed which has 53 layers of feature extraction convolutions and is deeper than Darknet-19. For YOLOv3 architecture, the standard input picture size is 416,416. The YOLOv4 architecture is made up of three primary blocks: the neck block, which adds layers between the head and the backbone CSPDarknet53, and CSPDarknet53 as the backbone (P. Jiang et al., 2022). A Path Aggregation Network (PANet) is used for feature aggregation in YOLOv4 to increase overall accuracy. Additionally, it employs Spatial Pyramid Pooling (SPP) blocks to divide core operations from important ones. Similar to YOLOv3, the head uses detection along with additional strategies like bags of goodies

and freebies to cut down on training and inference time which capture the input picture at a resolution of 608 by 608. Tiny-YOLOv4, on the other hand, receives the input picture with a resolution of 416,416. Studies reveal that YOLOv4 outperforms effective detection methods as the best real-time object detector without sacrificing accuracy. The most recent version YOLOv5, uses the PyTorch framework and has the same architecture (spine, stem, and output block) as YOLOv4. To extract features, YOLOv5 sends 640,640 resolution photos to the backbone and using Bottlenecks. By altering the height and width of the Bottleneck CSP, four variants (YOLOv5s, YOLOv5m, YOLOv5l, and YOLOv5x) may be produced. YOLOv4 scored better in mAP tests with COCO benchmark than YOLOv5, despite claims of improvements from his YOLOv4 as demonstrated in Figure 6 (Bochkovskiy et al., 2020).

Alexey Bochkovskiy et al, released YOLOv4 in April 2020. To make them more appropriate for single GPU training, they merged cross-iteration batch normalization and path aggregation networks. CSPDarknet, a hybrid of CSPNet and Darknet, served as the network's backbone, improving the learning capacity of convolutional layers while simultaneously lowering storage costs. To improve the pothole detecting process, the researcher additionally employed YOLOv4 algorithm for numerous object detection applications.

YOLOv4-based Pothole detecting techniques have been proposed. For the pothole detecting application, the author employed several modules from YOLOv4 and YOLOv5s. The study shows that YOLOv4-tiny is more accurate than YOLOv5. The primary disadvantage of this paper is that no parameters for determining how long it takes to identify potholes are supplied. This is due to the fact that it is a critical metric for real-time pothole identification. YOLOv4 was used to detect potholes in a paper by Omar and Kumar". Model accuracy and recall were determined and used to evaluate the model. However, the acquired results were lower than predicted (Omar & Kumar, 2020b).

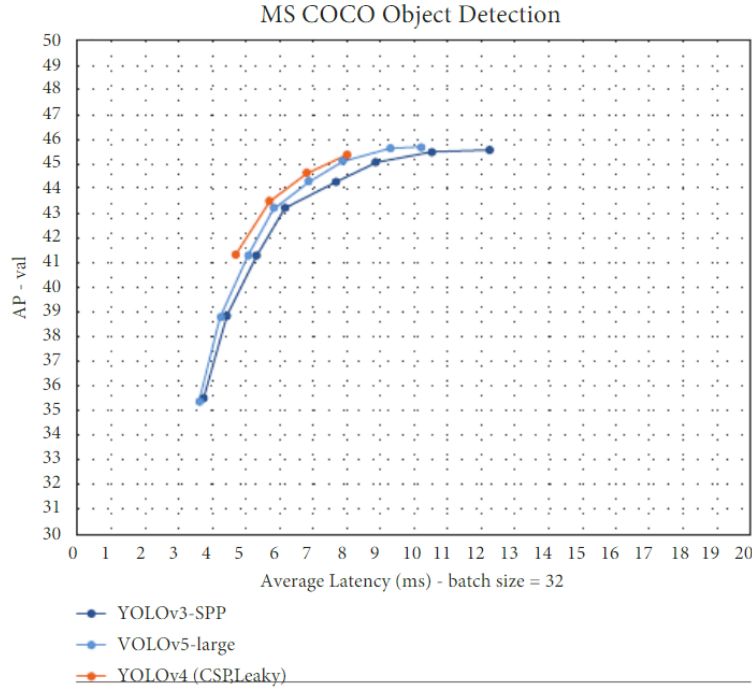


Figure 6: The performance of the YOLOv4 in compared to other models (Z. Jiang et al., 2020)

2.3 Review of Yolo Model for pothole detection

The researchers gathered 1,500 pictures of Indian roadways for their study. YOLO (You Only Look Once) is a dataset annotation and training tool. Data collection is also taught to YOLOv3, YOLOv2, and YOLOv3-tiny (J et al., 2020) (Mnasri et al., 2020)

The authors of this study analyzed the GPR data using the swept rectified power technique to evaluate the radar signal attenuation and depth curves. This article implements an innovative strategy for initializing neural network weights. To evenly distribute the data, a network is used with autoencoders trained on event related data only (Colagrande et al., 2020).

The Yolo neural network model was used to effectively identify potholes in images of asphalt pavement in a paper titled "Deep Learning Method Based on YOLO Neural Network for Detecting Potholes in Asphalt Pavement." It demonstrates that the appropriate Yolov3, Yolov3 Tiny, and Yolov3 SPP designs have adequate detection accuracy. These mAPs have accuracies in planimetry of 64.45%, 53.26%, and 72.10%, respectively, and have mAPs of 83.43%, 79.33%, and 88.93%. The average detection time per picture is 0.04s. It is therefore likely to be created and used. The YoloV3 algorithm was employed (Chavan et al., 2022b).

Apply the YOLOv4 framework to estimate objects and their locations easily and precisely in aerial images. The model's convergence to the VisDrone DET dataset for object recognition from the sky was sped up via transfer learning. On a Tesla K80 GPU,

the trained model produced an inference speed of 8.7 FPS and an average accuracy (mAP) of 45.64%, effectively recognizing clipped and occluded objects (Ali et al., 2021).

The latest stable releases of “YOLO, YOLOv3 and YOLOv4 show how the application has grown over time. The difference between the versions is that YOLOv3 uses a traditional pyramid network (FPN) to detect things, whereas YOLOv4 uses PANet to detect items as a way of aggregating parameters for different detection levels. With this advancement, YOLOv4 is much more accurate than YOLOv3, and different versions handle noise differently”. Potholes were also identified using a YOLO (You only look once) approach. Silva et al. evaluated road degradation using the YOLOv4 method using a custom dataset that provides aerial imagery of roads from flying drones. The accuracy and utility of the YOLOv4 algorithm in the context of recognizing damages on highways were experimentally evaluated, and a 95% accuracy was attained (W. Yang et al., 2020).

YOLOv3 is an item detection gadget that use the characteristic pyramid community (FPN) to hastily find gadgets in a special subject of view. Because of its ability to pick out and find tiny gadgets while not having to run numerous scans of the equal region, the gadget has had substantial achievement in locating them. One of the important things, improvements furnished with the aid of using this community is the addition of a brand-new classifier that lets in the gadget to comply with gadgets at diverse stages in their travels, permitting YOLOv3 to find smaller gadgets with greater success. However, the community suffers significantly in phrases of processing time. Various adjustments to the structure were performed to enhance the velocity of tiny object identity even greater (Darapaneni et al., 2021a).

We proposed optimizing the loss characteristic of the YOLOv3 by converting the default loss characteristic L2 and type loss cross-entropy to GIoU (generalized intersection-over-union) loss characteristic and focal loss respectively due to the lack of robustness of the L2 loss characteristic and vulnerabilities. Inclusive of the version being sensitive to examples with large mistakes and whilst attempting to adjust, sacrificing instance values with small mistakes. To that end, a GIoU loss characteristic model of the IoU loss characteristic is presented as a standard upgrade for the YOLOv3 network (J. Yang et al., 2010). Recommended network architectural adjustments to improve the processing speed and performance of YOLOv3. The problem of objects of different sizes was initially addressed by the authors by suggesting utilizing the K-means approach with the diameter and height of the object's bounding box to generate appropriate anchor boxes for the items of interest in a dataset. This innovation enables quicker network training since the produced anchor boxes are presently substantially closer to the dataset items (Darapaneni et al., 2021b).

Significant enhancements to the new edition include increased speed, power, and precision. This version included advanced methods such as batch normalization and

anchor boxes. Under the title "YOLOV3: Incremental Improvements," an updated version that was far better and more powerful than its predecessor was published in 2018. YOLOv3 surpasses cutting-edge detectors such as Retina Net, SSD, and variations, as seen in Figure 7. The graph shows a significant speed improvement over typical detection method (Sai et al., 2022).

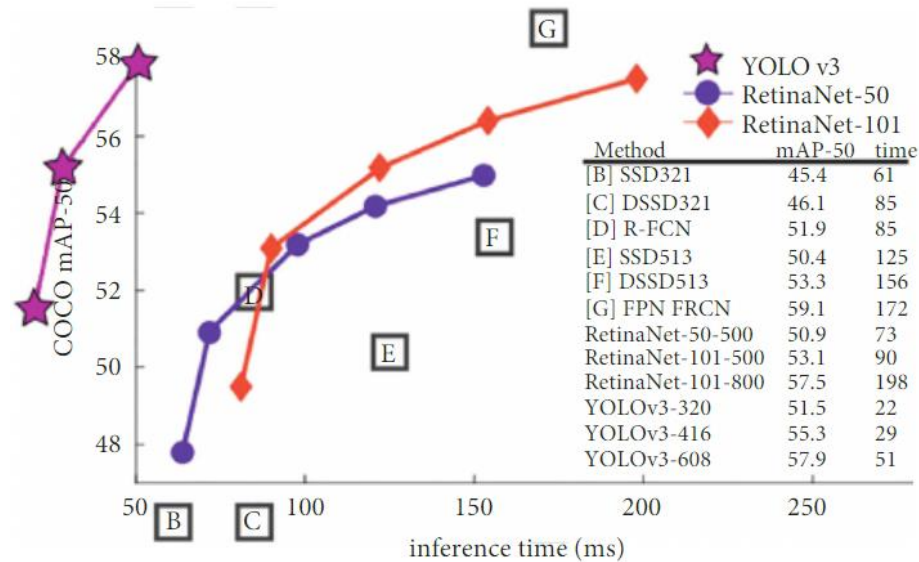


Figure 7: Comparing the detection model's performance to that of YOLOv3

“On April 23, 2020, Alexey Bochkovskiy, Chien-Yao Wang, and Hong-Yuan Mark Liao released YOLOv4, which promises to increase average accuracy by up to 10% and frames per second (fps) by 12% over YOLOv3. A reduced version of the original YOLOv4, a more user-friendly and quicker real-time object detector, is called Tiny YOLOv4”. Glenn Jocher's YOLOv5 was developed by Ultralights following the publication of YOLOv4. Keeping in mind that v5 is not a fork of the original, in contrast to Alexey's repository. They assert to be a lot lighter and quicker than YOLOv4. The accuracy, nevertheless, is equivalent to YOLOv4 (Khaliq et al., 2022).

The pothole detecting system employed YOLOv3, HOG, SSD, SVM, and Faster R-CNN, and the data set was gathered by attaching a smartphone to a moving car's dashboard. YOLOv3 surpassed its rivals by recognizing potholes and estimating their size in 1,500 photographs with 82% accuracy after being educated in each of the ways using machine learning and deep learning models. However, it was not considered an actual-world scenarios, as performance on extrapolated data is unsatisfactory (Ping et al., 2020b).

Road objects such as mileage signs, milestones, and other cars can be recognized, but the size of objects seen in a panoramic view varies greatly, making identification of small

objects far from the camera difficult. Panoramic cameras eliminate the need to see all directions with many cameras, but the image or video feed from a panoramic camera can be distorted or curved, reducing accuracy and interfering with the capture process.

2.4 Literature Review Findings

Section II focused on research that used the Internet of Things (IoT), GPS systems, ground penetrating radar (GPR), acoustic road monitoring, and deep learning techniques/algorithms to identify road degradation. The researchers determined that deep learning-based image capture devices are the most successful and cost-effective way for identifying road damage, particularly potholes, after reviewing the aforementioned methodologies. Using YOLOv4's deep learning-based architecture, a model for identifying potholes is constructed in this study.

3. Proposed Methodology

3.1 Dataset Description

Utilized a labelled image dataset with more than 300 photographs of roads with potholes for this study. This dataset is from Kaggle and is in the public domain under the CC0 license. Created an image file with annotations for each image in the collection. In this approach, when the deep learning model is being trained to recognize the location or position of objects in the image, information is supplied alongside the image. Consider imlabel after labeling with any tool. Data flows are converted into recognized record formats. Learning is accelerated because it is a binary format that is very fast to read and write. This greatly speeds up training.



Figure 8: Samples images of Dataset

3.2 Proposed System's Technical Components

A technical component is the part of a procedure or service that involves setting up equipment and technician time.

3.2.1 Requirements for Hardware

A decent laptop with a Nvidia 1660 GPU and 8 GB of RAM was enough, as YOLO works best with a high-quality GPU.

3.2.2 Software Requirements

Google Collab

Collaborator was created by Google Research. Colab excels in three areas: data analytics, training, and machine learning. Anyone with a browser can write and run arbitrary Python code. “Colab is a hosted Jupyter notebook service that provides free access to computing resources such as GPUs. Collaborator was created by Google Research.

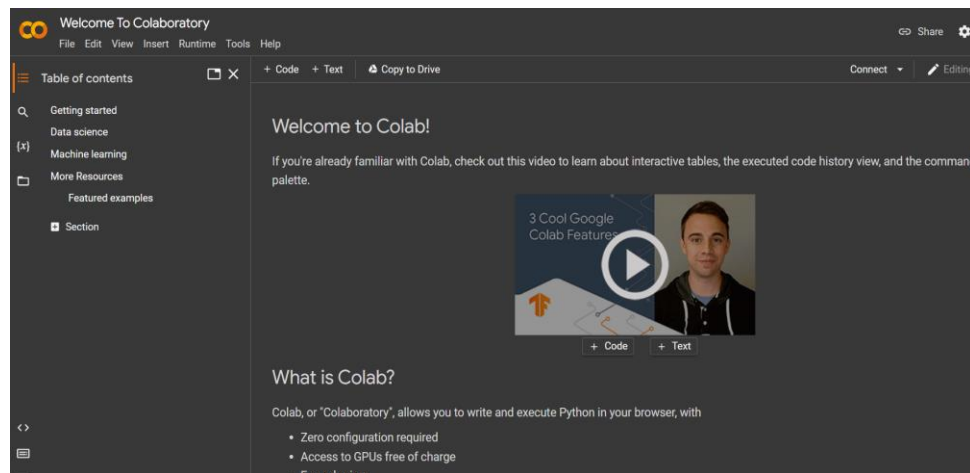


Figure 9: Interface of google Collab

Jupyter Notebook

Jupyter is an interactive development platform for code, data, and notebooks. It is an open-source interactive development environment. Users can configure and arrange data science, scientific computing, and machine learning workflows using the interface's flexibility.

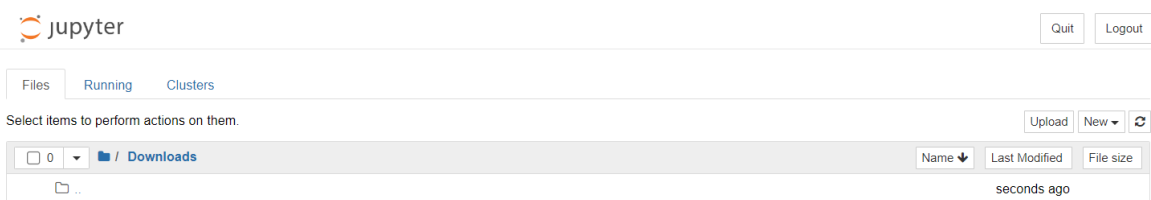


Figure 10: Interface of Jupyter

3.3 Implementation

A major priority for UK telecommunications services is enhancing road safety. This suggested model builds a pothole detection system using deep learning and image processing, both of which leverage the YOLO technique. This technique helps to alert vehicles of potholes on the road.

3.4 Image Preprocessing

A key step in the data mining process is data preprocessing which involves modifying, deleting, or adding data before use to ensure or improve performance. Garbage in - garbage out is especially true for data mining and machine learning projects. A real-time detection system automatically extracts images from live camera footage and further processes them with the YOLOv4 algorithm. YOLOv4 models are pre-trained, so previously we had to input photos or videos as data.

3.4.1 Label Image

Labellmg is a free open-source program for visually labeling images. It has a graphical interface developed in Python and integrated with QT. It's a quick and easy approach to identifying hundreds of photos for your next object detection project. Labels are used in unlabeled datasets to indicate which data components we want in the model to recognize. Computer vision and developing powerful models require high-quality data sets. The creation of computer vision models follows the principle of garbage in, garbage out. However, it is important to label photos carefully and accurately. To ensure the quality of the training data set, we created a labeling guide.



Figure 9: Annotation done on images using labeling

Each dataset must be prepared as a TXT file and stored with the JPEG picture before training in order to build bounding boxes for the indentations using the annotation library as shown in the above figure.

3.5 YOLOv4 Algorithm

Object recognition techniques have matured since the advent of CNNs and R-CNNs a few years ago, but competition remains fierce. Most CNN-based object detectors are only useful for recommender systems. For example, when using a video camera, a step-by-

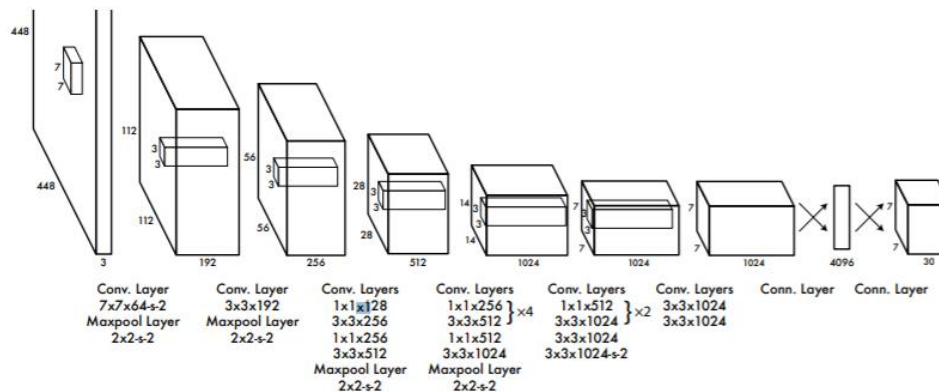


Figure 10: Detection Network

step, accurate model is used to find an open parking space. YOLO often called You Only Look Once is an object recognition system that detects many things in one housing in real time. It also recognizes objects faster and more accurately than other recognition systems. Up to 9000 types of visible and invisible objects can be estimated. A real-time detection system can be quickly trained and applied to a production system. You can set a limited border around adjacent items to distinguish many from one photo. Here the images are divided into regions and imagine boxes with bounding edges and probabilities for each region, based entirely on convolutional neural networks (CNN). In addition, it provides various bounding edge boxes and probabilities for these classes. Through, training and testing, YOLO observes the big picture and secretly encodes classes of contextual information and their properties. Modeled after the GoogleNet design, YOLO's architecture consists of a total of 24 convolutional layers, with two fully connected layers at the end (Omar & Kumar, 2020b).

Bochkovskiy et al., recommended YOLOv4 as an improvement over YOLOv3. The method performs best on the Tesla v100 GPU, yielding results with an average accuracy of 43.5% and 65 frames per second. “As methodological changes to modern methods of regularization and data augmentation, YOLOv4 proposes the addition of weighted residual connections, cross-minibatch normalization, cross-stage partial connections, self-adversarial training”. The author also adds that YOLOv4 Tiny version which sacrifices prediction accuracy but offers faster object detection and higher frame rate (Bochkovskiy et al., 2020).

Here's a timeline showing how YOLO has changed over time.



Figure 11: Yolo timeline

The YOLOv4 algorithm is employed by the suggested system. Joseph Redmon created a cutting-edge real-time object recognition system that can recognize several items in a single frame. Compared to YOLO prior technologies, it employs a totally new detecting method. Utilize a single neural network to process the entire picture. This network divides the picture into regions, and then forecasts the bounding boxes and probabilities for each zone. These bounding boxes are weighted based on predicted probability (Prof. S. M. Patil et al., 2022).

YOLOv4 is doing well with AP and FPS improvements. YOLOv4 improves real-time object detection and training on a single CPU. YOLOv4 on Tesla V100 achieved peak performance at 43.5 percent speed (AP) at 65 frames per second (FPS) on the COCO dataset. The diagram below shows how YOLO works in its most basic form. Each raster cell in the $S \times S$ raster that YOLO creates from the input image predicts which element will be centered there.

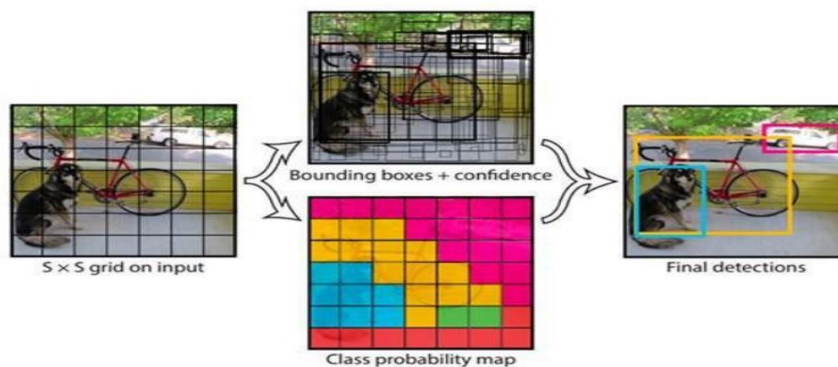


Figure 12: $S \times S$ grid (Sai et al., 2022)

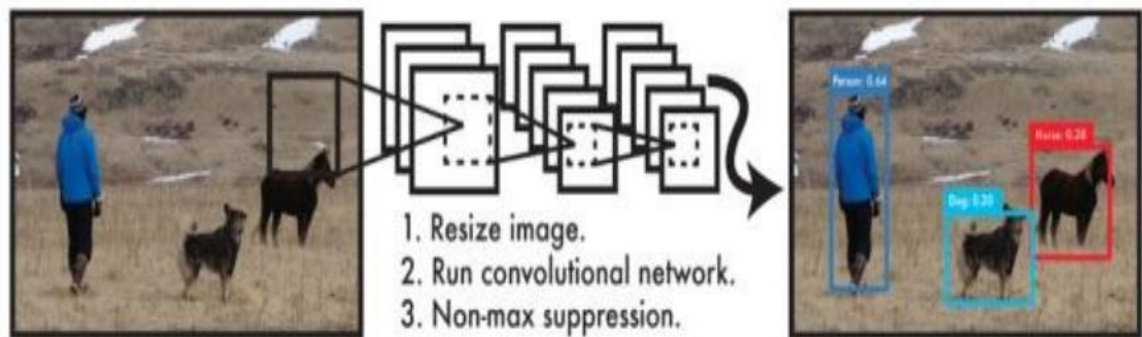


Figure 13: Concept of Yolo (Sai et al., 2022)

The bounding boxes and confidence intervals for these boxes are as follows: each grid cell's prediction, confidence intervals, it symbolizes the model's conviction that it is contained inside a box, accuracy, and objects. They think the box contains the forecast (Sai et al., 2022).

3.5.1 Using Darknet and YOLOv4 to find potholes

Using the YOLOv4 object detection methodology and the darknet framework, creating a pothole detecting system. The Darknet project is an open-source object detection framework that provides YOLO model training and inference support. C is used to write this library.



Figure 14: Darknet

Joseph Redmon launched the Dark Web Project in 2014 after the first YOLO paper was released. It was purchased by Alexey Bochkovskiy”, who runs an active branch of the

original repository, not long after the release of YOLOv3. The best object recognition model is YOLOv4 model, was also given support.

From the dataset we'll divide few images to train, to validate and finally to test. Rebuilding Darknet from the ground up by cloning the Darknet development. Run the following command from the terminal.

1) Clone, configure & compile Darknet

```
# Clone
!git clone https://github.com/AlexeyAB/darknet

Cloning into 'darknet'...
remote: Enumerating objects: 15478, done.
remote: Counting objects: 100% (18/18), done.
remote: Compressing objects: 100% (8/8), done.
remote: Total 15478 (delta 9), reused 14 (delta 8), pack-reused 15460
Receiving objects: 100% (15478/15478), 14.06 MiB | 21.36 MiB/s, done.
Resolving deltas: 100% (10397/10397), done.
```

Darknet YOLOv4 training and testing requires all image paths in a text file. These text files are used for image path mapping. Darknet YOLOv4 training and testing requires all image paths to be text files. These text files are then used to map image paths.

6) Create train.txt file

```
import glob
images_list = glob.glob("data/obj/*.jpg")
with open("data/train.txt", "w") as f:
    f.write("\n".join(images_list))
```

We employ an existing, pre-trained tiny model to train the new model. Run the following command in the terminal to download it:

7) Download pre-trained weights for the convolutional layers file

```
#!wget https://pjreddie.com/media/files/yolov4.conv.137
# Download the yolov4 pre-trained weights file
!wget https://github.com/AlexeyAB/darknet/releases/download/darknet_yolo_v3_optimal/yolov4.conv.137

--2022-08-14 22:28:04-- https://github.com/AlexeyAB/darknet/releases/download/darknet_yolo_v3_optimal/yolov4.conv.137
Resolving github.com (github.com)... 192.30.255.113
Connecting to github.com (github.com)[192.30.255.113]:443... connected.
HTTP request sent, awaiting response... 302 Found
Location: https://objects.githubusercontent.com/github-production-release-asset-2e65be/75388965/48bfe500-889d-11ea-819e-c4d18
--2022-08-14 22:28:04-- https://objects.githubusercontent.com/github-production-release-asset-2e65be/75388965/48bfe500-889d-
Resolving objects.githubusercontent.com (objects.githubusercontent.com)... 185.199.108.133, 185.199.111.133, 185.199.109.133,
Connecting to objects.githubusercontent.com (objects.githubusercontent.com)[185.199.108.133]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 170038676 (162M) [application/octet-stream]
Saving to: 'yolov4.conv.137'
```

Running inference after training gave us a reasonable sense that using deep learning to tackle real-world problems might be harder than it looks. Different results from different models made it pretty clear.

3.5.2 Configuration of YoloV4

It is difficult to set up the training configuration for YOLOv4 for a custom dataset. The learning rate and batch size will be set. The configuration is created by combining a number of chunked configuration files. Using YOLOv4, take the following actions.

- Set the stack size to 64. Batch size determines the number of photos per iteration.
- Set tiling to 16. Tiling is the number of pieces the GPU memory stack is divided into.
- “max_batches to 2000 * number of classes”
- Steps to 80% and 90% of max batches.
- Change num_classes in all of the YOLO layers.
- Change filters in all of the YOLO layers.

3.5.3 System Flow

To acquire the required result from this model, image/video should be submitted to the code. The system flow is as follows: when the code is activated it starts the process and detects potholes in real time using the YOLOv4 algorithm. Potholes that have been detected are presented in a bounding box, allowing for real-time pothole detection.

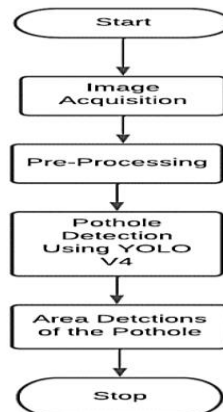


Figure 15: Wrok flow diagram

3.5.4 YoloV4 Evaluation

Mean Accuracy (mAP) is a metric used to assess object detection algorithms like R-CNN and YOLO. Mean Accuracy (mAP) computes a score by comparing the detected bounding boxes to the ground truth bounding boxes. The higher the score, the more accurate the model's detection.

Mean Accuracy (mAP) is a typical assessment metric for object detection in computer vision (that is, localization and classification). Localization establishes where the instance is (e.g., bounding box coordinates), whereas categorization describes what it is. Accuracy gauges how accurately a forecast was made. That is, the proportion of projections that were accurate. Count the number of the model's predictions that came true.

$$\textbf{Precision} = \frac{\textbf{TP}}{\textbf{TP} + \textbf{FP}}$$

TP = True Positives (Predicted as positive as was correct)

FP = False Positives (Predicted as positive but was incorrect)

If that was the case, let's calculate mAP for the following image:

From the image, we get:

$$\textit{True Positives (TP)} = 1$$

$$\textit{False Positives (FP)} = 0$$

$$\textbf{Precision} = \frac{\textbf{TP}}{\textbf{TP} + \textbf{FP}} = \frac{\textbf{1}}{\textbf{1} + \textbf{0}} = \textbf{1}$$

According to Table 1, a true positive (TP) is the precise identification of an object in an image. An erroneously identified item is known as a false positive (FP). Network recognizes objects that are hidden from view. A True Negative (TN) is the correct identification of an object that does not appear in the image, as opposed to a False Negative (FN), which is an object that appears in the image but is not recognized by the network.

Table 1: Table 1. Unnormalized confusion matrix.

Actual/Prediction	Predicted as Positive	Predicted as Negative
Positive	TP	FN
Negative	FP	TN

“The ratio of true positives (TP) to true positives (TP + FN) is the definition of recall, while the ratio of true positives (TP) to the total number of positives (TP+FP) is the definition of precision. The accuracy and memory parameters are two fractions with the same numerator but distinct denominators, according to mathematics. Additionally, non-negative numbers less than or equal to 1, precision and recall”.

“A high accuracy of 14 denotes the correctness of the things discovered. A high True Positive rate corresponds to a strong recall. H. Positive things have a low rate of missingness. By adjusting the threshold and assessing the accuracy and recall values, the model is assessed. The computations for precision and recall make use of N thresholds, with each threshold standing for a pair of precision (P_n) and recall (R_n) values (n = 1, 2, ..., N). In equation, average precision (AP) is defined” (Lin et al., 2021).

$$AP = \sum_{n=1}^N (R_n - R_{n-1}) P_n$$

Set mAP to 0.5 to compare the effectiveness of the three models. TensorFlow is used to implement the YOLOv4 model. The model training maturity was evaluated stepwise by different iterations and image resolutions. To determine whether the detection is accurate or inaccurate, the Intersection Over Union (IoU), which calculates the amount of overlap between the predicted bounding box and the ground truth bounding box of the actual element, is compared to a threshold gain. The Average Accuracy (AP) metric value depends on the threshold, so we have to choose one.

In this study, the threshold was established at 0.3. For detection boxes to be deemed effective, the IoU must be at least 30%. Predictions produced with thresholds between 0.5 and 0.3 are widely accepted to be indistinguishable by the human eye. In this study, the threshold was established at 0.3. Therefore, box detection is only considered to be effective if the IoU is 30% or above. It is commonly believed that predictions produced with thresholds between 0.5 and 0.3 may be invisible to the human eye. The threshold was set at 0.3 in this investigation. As a result, box detection is only regarded effective if the IoU is 30% or above. For each prediction, the current state in object recognition generates thousands of "anchor fields" or "previous fields" that indicate the ideal position, shape, and size of an item dedicated to the prediction. Intersection Over Union (IoU) determine which bound field shows which feature border field has the most duplicates divided by the fewest duplicates of each bound field and find out which element gives the largest IoU if the maximum value is greater than 50%. This will be reported to the neural network as True. If the IoU is greater than 40%, learn nothing from this example and ensure that the object does not exist if the maximum IoU is 40%. Existing pieces of art perform well in this way, and the thousands of predictors do a great job of predicting whether their type of object will appear in an image. The default anchor box

configuration might provide highly specialized predictors, and it's possible that some items in the image won't achieve 50% with any anchor box. In this case, the neural network won't ever be aware that these things exist and won't ever have the ability to predict them. It is necessary to set a threshold since the value of the mean accuracy index (AP) depends on it. The threshold in this investigation was set at 0.3. Since the IoU must be greater than or equal to 30% in order for the detection box to be legitimate.

4. Test and Results

We analyze and evaluate the results of this section on feature domains, classifiers, and datasets. The specifics and parameters of the method have been described in the previous section.

YOLO v4 Model	To detect potholes, we employed the YOLOv4 image categorization system. By providing photos or videos to the code, the algorithm identifies potholes in real time. It was a pre-trained algorithm that could detect potholes in real time.	In real time, the model readily reached outstanding accuracy, averaging 85% to 90%. System for detecting potholes.
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4.1 Test Output

A bounding box is a boundary that shows the anticipated location of an item. There are five attributes for each prediction item that resides inside a bounding box (i.e., x, y, w, h, confidence). In regard to the placement of the box within the cell, the box's center coordinates are x and y. Since the cell is not in control of the center if it is outside of it, it cannot represent it. One reference to the object in the center appears in each cell consistent with these sites. The box's width and height, indicated by the letters w and h, are also scaled to the size of the image.

A bounding box is used to show whether there are mistakes or not, with probability values reflecting levels of confidence (c_{ji}). Equation provides the bounding box's confidence value (5). The bounding box is displayed if the bounding box's J-value exceeds the threshold. If not, they are gone.

$$C_i^j = P_{i,j} * IoU_{pred}^{truth} \quad (5)$$

According to the figure it demonstrates the detection output for the road surface flaws identified by the YOLOv4 model. In comparison to the YOLOv4 model, more thoroughly, without omission, and with a higher confidence value identifies faults on the road surface.



Figure 16: Detected pothole Result Image (1), 75 to 95 % accuracy achieved



Figure 17: Detected pothole Result Image (2,3), 75 to 95 % accuracy achieved



4.2 Performance of YOLOv4

For YOLOv4, a pretrained weight model and suitable filters in convolutional layers were employed in the training process. Every 1000 iterations, the model weight parameters are stored. The loss function of the YOLOv4 model is divided into three parts: class loss, location loss, and confidence loss. The YOLOv4 training procedure, as shown in the image, runs over 4000 epochs and takes over 7 hours on the data set. The training process looks to be steady, with a 77.7% success rate at mAP 0.5. The model training procedure does in fact require substantially less time than the YOLOv4 model training approach. The average success value of 78.7% is somewhat higher than in YOLOv4 model with the same mAP 0.5, indicating that the training process is more reliable. The afocal loss function, on the other hand, solves class imbalance in object recognition training.

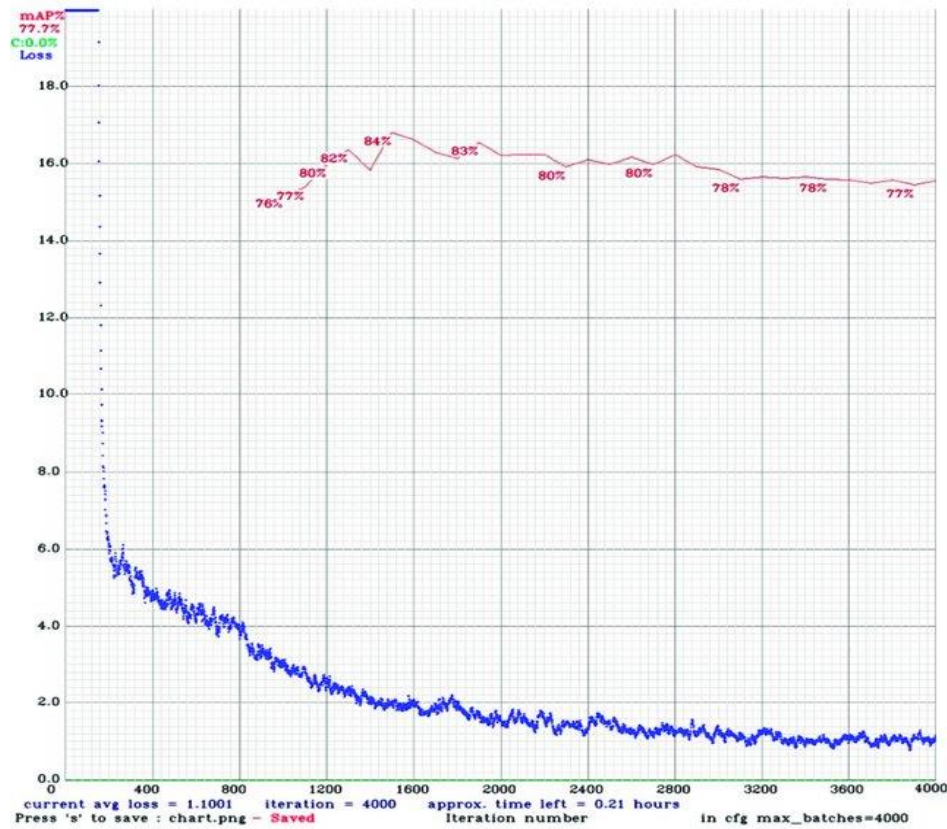


Figure 18: YOLOv4 outputs acquired after 4000 epochs using 720 720 pictures

Focus lapse to focus training on hard negative instances, the modulation principle is used to cross-entropy loss. This is a dynamically scaled cross-entropy loss in which the scaling factor approaches 0 as confidence in the proper class grows. Intuitively, this scaling factor minimizes the contribution of simple cases during training, allowing the model to focus on challenging examples more rapidly.

A loss function that classifies the input data points in the dataset computes predictor performance. The lower the loss number, the more accurately the classifier predicts the connection between the input data and the output target. The figures representing the progressive learning process for YOLOv4 and YOLOv4-tiny show the gradual decay of the loss value after each epoch. Both YOLO loss functions were used to generate the curves. The curve validates the YOLOv4 model's training history.

Table 2: the mAP_0.5 value of YOLOv4

Model	mAP(%)
YoloV4	77.7%

Table 3 displays the precision and recall levels. It can be noticed that the recall value of YOLOv4 (74%) is higher than the precision value (84%).

Table 3: Precision and Recall of Model

Model	Precision	Recall
YoloV4	84	74

4.3 Conclusion

In this work, we use YOLOv4 models to identify pothole dirt in photographs of the road surface. The dataset, which was used to train the model, has a higher average accuracy than YOLOv4 (78.7%), which is useful for applications involving pothole identification. To increase pothole detection accuracy, it would be desired to further enhance the model by extending the backbone network architecture in detail. Furthermore, automating the pothole identification technique in further experiments can increase the execution efficiency of the YOLO model. To discover potholes, obtain new insights, and establish the most effective approach for the problem, this study employs experimental analysis and direct application of the YOLO series. In fact, the best model in this study may aid in making more accurate predictions in future investigations. Furthermore, minor indentations observed at a distance are less precise. The effects of severe weather and a lack of light were not examined. These limitations may be addressed in future investigations.

5. Conclusion and Discussion

5.1 Conclusions drawn from the data modelling

This study presented cutting-edge deep learning models for real-time pothole recognition (YOLO family and SSD-mobilenetv2) with the aim of deploying them on edge devices. It was discovered that the YOLOv4 model has the best fit for accuracy, while the Tiny-YOLOv4 model has the simplest fit for real-time pothole detection, with 90% detection

accuracy and 31.76 FPS. The proposed method can help road maintenance authorities establish speedy and effective road infrastructure repair procedures. A more advanced method, using the global positioning system "GPS", helps identify and locate defects in the pavement. This algorithm has the potential to benefit the automation industry. This survey can be extended to detect various pavement and road indentation problems, classify roads by quality, and estimate pothole depth. Accuracy limitations may be addressed in the future by further modifying and extending the real-time implementation.

5.2 Strength of the Study

The incidents of asphalt pavement have been and are being concerned by the authorities to avoid unexpected cases. These pavements are susceptible to scenarios such as traffic loads, weather conditions, age, poor materials used for construction, and poor drainage, revealing two major pavement faults such as: cracks and potholes. Potholes are basically denting in the road surface that need attention because they cause terrible situations like accidents, unpleasant driving experiences and faulty vehicles as shown in Figure 1, which need to be addressed first to tackle potholes to minimize their impact. According to the WHO (World Health Organization), road traffic accidents will become the 5th leading cause of death by 2030.

On the dataset, the YOLOv4 architecture produces the best results in terms of speed and accuracy. In real-time object recognition applications, a one-step object detection model with fast inference can be used. As a result, even tiny drones can be equipped with high-level object detection technology. With more computing resources, the model can be scaled for greater resolution and network size, increasing accuracy even further. Additionally, training the model with higher quality photos can further improve accuracy.

5.3 Limitations of the Study

The limitations of this study must be addressed in future research. The training dataset is relatively tiny, which causes two issues. For starters, it limits model selection; for example, the neural network was rejected since it required a big dataset to create features on its own. Second, it restricts the model's accuracy improvement. Through the application of swarm intelligence to aggregate data, community-sourced data will enable major system improvement. Another issue with the dataset was the unequal distribution of samples across the three categories, which caused biases that might influence the classifier's accuracy and individual recall. Because of the small number of samples, "cross-sectional" pavements are not properly characterized. The vehicle's performance is affected by data gathering parameters such as vehicle type, cargo, travel speed, smartphone position, and location in the vehicle where the smartphone is situated. Incorporating a vision-based component to improve the vibration detection technique is one prospective topic that might be studied to improve the vibration detection method. Because the acquired data is related to user private information (e.g., driving trajectories,

video footage), some encryption and anonymization solutions will be necessary. To summaries, all of the constraints described above are possible study subjects that should be addressed in future studies.

5.4 Future Research

We intend to improve the model by deepening the backbone network architecture for improved pothole detecting accuracy. Furthermore, in a future study, the efficiency of running the YOLO model might be increased by automating the pothole labelling technique. This work uses the YOLO series directly to the pothole identification problem in order to use fresh information as well as experimental analysis for comparison, identifying the optimum model for the pothole detection problem. Furthermore, minor potholes placed at a distance have poor accuracy. There was no research on adverse weather or low light circumstances. These constraints might be addressed in a future investigation.

5.5 Overall Conclusions

Pothole identification is tough and unusual because, unlike other objects, they do not have a defined shape or size. They appear in a variety of shapes, making identification considerably more challenging. Average precision, average recall value of the model is calculated, and all parameters are displayed. The dataset used to train the model contains photographs of potholes taken in various ways and many potholes.

The decision to use YOLOv4 is great because the main advantage of adopting YOLO is its incredible speed; it can handle 45 frames per second. YOLO is also aware of the general representation of objects. This system has various advantages and can be used with fewer people. As a result, it did a good job training and testing our model using YOLOv4. Pothole detection technology is effective with approximately 95% accuracy.

The findings of the investigation suggest that the YOLOv4 model is ideal for pothole detection. This is due to the model's tiny size and low computing power requirements, allowing it to be quickly deployed with less storage space and reduced power consumption. In comparison to other models, the accuracy is likewise extremely good. The inference speed of the model is slower than that of other models, but the accuracy-speed trade-off favors YOLOv4 model. As a result, the suggested model can identify potholes more accurately and is ideally suited for real-time applications.

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